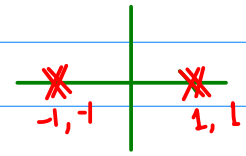
L37

19/11/2020

$$A = \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



$$H = \begin{bmatrix} A & M \\ -Q & -A^T \end{bmatrix} \qquad \text{spec}(H) = \{-1, -1, 1, 1\}$$

$Hv_2 = -v_2 + v_1$

$\Rightarrow (H+I)v_2 - v_1 = 0$

linsolve$(H+I, -v_1)$

$Ax + b = 0$

Stabilizing Solution & Riccati Operator:

Def^n: Suppose H is a Hamiltonian matrix. Then H is said to be in the domain of the Riccati operator if $\exists$ $n \times n$ matrices H_-, X_1, X_2 such that

$$H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \quad \text{with } H_- \text{ being Hurwitz \& } \boxed{X_1} \text{ being invertible.}$$

set of Hamiltonian matrices

WLOG

$$H \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} H_-$$

$$\text{Ric} : \mathcal{H} \longrightarrow \mathcal{X}$$

$$\text{Ric}(H) = X := X_2 X_1^{-1}$$

$$\text{Ric}(H) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

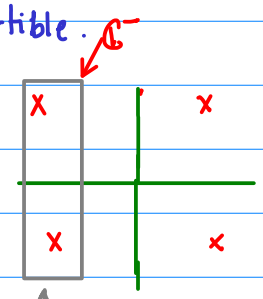
$\Downarrow$

$\text{spec}(H) \to \mathbb{C}^+, \mathbb{C}^-$

$2n \times 2n$

$2n$ eigenvalues

$\mathbb{C}^+$ $\mathbb{C}^-$



eigenvectors

$\text{spec}(H) = \{-1, -1, 1, 1\}$

2-dim $\mathbb{R}^4$ $\quad X_2 X_1^{-1} = 0$

Theorem: Let H belong to the domain of the Riccati operator and $X = \text{Ric}(H)$. Then

(1) $X_2 X_1^{-1}$ is symmetric (X is symmetric)

✓ (2) X satisfies $A^T X + XA + XMX + Q = 0$

(3) $(A + MX)$ is Hurwitz

Proof: $\because X = \text{Ric}(H) \Rightarrow \boxed{H \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} H_-}$ Hurwitz

$$J := \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$$

$$JH = (JH)^*$$

$$\Rightarrow \underbrace{\begin{bmatrix} I & X^* \end{bmatrix} JH \begin{bmatrix} I \\ X \end{bmatrix}}_{\text{symmetric}} = \underbrace{\begin{bmatrix} I & X^* \end{bmatrix} J \begin{bmatrix} I \\ X \end{bmatrix}}_{} H_-$$

$$\therefore \begin{bmatrix} I & X^* \end{bmatrix} J \begin{bmatrix} I \\ X \end{bmatrix} H_- = \left(\begin{bmatrix} I & X^* \end{bmatrix} J \begin{bmatrix} I \\ X \end{bmatrix} H_- \right)^* = -H_-^* \begin{bmatrix} I & X^* \end{bmatrix} J \begin{bmatrix} I \\ X \end{bmatrix}$$

$$J^* = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} = -J$$

$$\Rightarrow \begin{bmatrix} I & X^* \end{bmatrix} \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} H_- = -H_-^* \begin{bmatrix} I & X^* \end{bmatrix} \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix}$$

known known

$$\boxed{A^T P + PA + Q = 0}$$

Unknown

$$\Rightarrow (X^* - X) H_- = -H_-^* (X^* - X) \Rightarrow \boxed{H_-^* (X^* - X) + (X^* - X) H_- = 0}$$

H_- is Hurwitz $\Rightarrow (X^* - X)$ is a unique solution

$X^* - X = 0 \Rightarrow X^* = X$

A is Hurwitz then P is unique

(c)

$$H \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} H_-$$

$$\Rightarrow \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} A & M \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} H_-$$

$$\Rightarrow A + MX = H_-$$

$X = Ric(H)$

$$\boxed{H \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} (A + MX)}$$

$M \in \mathbb{R}^{n \times n}$

$M \geq 0$, $\forall x \in \mathbb{R}^n$, $x^T M x \geq 0$ } +ve semi def^n

$\Downarrow$

eigenvalues of are all in $\bar{\mathbb{C}}^+$

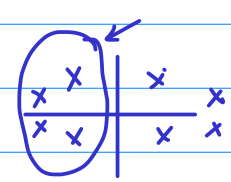
Theorem :- Suppose H is Hamiltonian matrix. Assume

(a) $spec(H) \cap j\mathbb{R} = \phi$ ← no imaginary axis eigenvalues

(b) M is semidefinite ($M \geq 0$ / $M \leq 0$)

(c) (A, M) is a stabilizing matrix pair.

Then H is in the domain of the Riccati operator.



Proof: $\because spec(H) \cap j\mathbb{R} = \phi$

$$H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_-$$

Show X_1 ← Invertible

Step A: $X_1^* X_2$ is symmetric

$$[X_1^* \;\; X_2^*]\, JH \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \underbrace{[X_1^* \;\; X_2^*]\, J \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_-}$$

$\because JH$ is symmetric

$$[X_1^* \;\; X_2^*] \begin{bmatrix} -X_2 \\ X_1 \end{bmatrix} H_- = (X_2^* X_1 - X_1^* X_2) H_-$$

$$\|$$

$$H_-^* [X_1^* \;\; X_2^*]\, J^* \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = -H_-^* (-X_1^* X_2 + X_2^* X_1)$$

$$(X_2^* X_1 - X_1^* X_2) H_- + H_-^* (X_2^* X_1 - X_1^* X_2) = 0$$

$\because H_-$ is Hurwitz $\Rightarrow X_2^* X_1 - X_1^* X_2 = 0$

$\Downarrow$

$X_1^* X_2$ is symmetric.

Step B:

$$[I \;\; 0]\, H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [I \;\; 0] \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \Rightarrow AX_1 + MX_2 = X_1 H_-$$

Let $v \in \ker X_1$, $\quad AX_1 v + MX_2 v = X_1 H_- v \Rightarrow MX_2 v = X_1 H_- v \longrightarrow (1)$

$$x^* X_2^* (AX_1 + MX_2) x = x^* \underbrace{X_2^* X_1}_{0} H_- x$$

$$\Rightarrow x^* X_2^* M X_2 x = x^* \cancel{X_1^* X_2} H_- x$$

$\because x^* X_2^* M X_2 x = 0$

$\Downarrow$

$\because M$ is semidefinite $\Rightarrow MX_2 x = 0 \longrightarrow (2)$

$\Rightarrow x^* X_2^* M = 0$

$X_1 H_- v = 0$

$\Downarrow$

$\ker X_1$ is H_- invariant

$\exists\, x \in \ker X_1$ such $H_- x = \lambda x$

$$[0 \;\; I]\, H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [0 \;\; I] \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \qquad H = \begin{bmatrix} A & M \\ -Q & -A^* \end{bmatrix}$$

$$\Rightarrow (-QX_1 - A^* X_2) = X_2 H_-$$

$$\Rightarrow (-QX_1 - A^* X_2) x = X_2 H_- x$$

$$\Rightarrow -A^* X_2 x = \lambda X_2 x \Rightarrow (\lambda I + A^*) X_2 x = 0 \longrightarrow (3)$$

$$x^* X_2^* [A + \lambda^* I \;\; M] = 0$$

$\because (A, M)$ is stabilizable $\Rightarrow x^* X_2^* = 0$

$\Rightarrow X_2 x = 0$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} x = 0 \quad \times \text{ Contradiction} \qquad \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \to \text{invariant}$$

full-column rank

$$\underline{\underline{x = 0}}$$

$\therefore \ \ker X_1 = \{0\} \to X_1$ is invertible

Corollary:

$$\dot{x} = Ax + Bu \qquad y = Cx$$

$$H := \begin{bmatrix} A & -BB^T \\ -C^TC & -A^T \end{bmatrix}$$

with A, B stabilizable & (C, A) detectable.

The H belongs to the domain of the Riccati operator.